

More than 67.302% of the zeros of the Riemann zeta function are simple and lie on the critical line

Abstract

Let $N(T, 2T)$ count the zeros of the Riemann zeta function with ordinates in $(T, 2T]$, with multiplicity, and let $N_0^s(T, 2T)$ count the simple zeros on the critical line in that interval. We refine the rank–trace argument of Claude [1] and prove

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.673025467453 \dots$$

The proof uses the analytic estimates of Theorem D in [1] and a rigorously verified seven-point inequality for the simple-zero vectors.

1 Statement and setup

We use the notation and normalisations of [1, §§2,4,7.1]. Put

$$\ell := \log \frac{T}{2\pi}, \quad I := (T, 2T], \quad D_0 := T^{1/2}, \quad I' := (T - D_0, 2T + D_0].$$

For the optimised test family of Theorem D, $L = \ell$ and

$$\phi(u) = \cos\left(\frac{\sqrt{2}u}{\ell}\right)^{1/2} \varrho(L/2 - |u|),$$

where the fixed-width ramp ϱ is defined in [1, §2.2]. Let

$$a := \frac{1}{L} \int_{\mathbb{R}} \phi(u)^2 du, \quad \widehat{G} := \frac{G}{aL^2}, \quad \widehat{A} := \frac{A}{aL^2}, \quad d := \left\lfloor \frac{LT}{2\pi} \right\rfloor.$$

As in [1, §4.1], divide the zeros in I' into the simple critical-line zeros $\mathcal{S}_1$, the distinct multiple critical-line zeros $\mathcal{S}_2$, and the unordered off-line pairs $\mathcal{P}$. Write

$$s_1 := \#\mathcal{S}_1, \quad s_2 := \#\mathcal{S}_2, \quad p := \#\mathcal{P}.$$

Then

$$N(I') \geq s_1 + 2s_2 + 2p. \tag{1}$$

For $\rho \in \mathcal{S}_1$, define

$$u_\rho := (\widehat{\phi}(\gamma_\rho - \tau_k))_{0 \leq k < d} \in \mathbb{R}^d, \quad v_\rho := \frac{u_\rho}{\sqrt{a}L}.$$

The simple-zero part of $\widehat{A}$ is

$$P_1 = VV^\top = \sum_{\rho \in \mathcal{S}_1} v_\rho v_\rho^\top,$$

where the columns of V are the v_ρ . By Lemma 2.2 of [1], $\|v_\rho\| \leq 1$. For

$$Q' := \hat{A} - P_1,$$

the decomposition in the proof of Proposition 4.4 of [1] gives

$$n_+(Q') \leq s_2 + p. \quad (2)$$

Set

$$H_{\text{MT}} := \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 0.6725007036794116457 \dots$$

Theorem D of [1] proves the lower bound H_{MT} . We prove the following refinement.

Theorem 1.1. *One has*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{267 \cdot 10^9 H_{\text{MT}} - 5.32 \cdot 10^8}{266,001,357,363} = 0.6730254674531403414 \dots$$

2 A stability refinement of the rank–trace inequality

Define the convex function

$$\Psi(t) := \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t-3, & t \geq 2. \end{cases} \quad (3)$$

For a positive semidefinite matrix M , let

$$\mathcal{D}(M) := \text{tr } \Psi(M).$$

Lemma 2.1 (Stability-enhanced rank–trace inequality). *Let V be a $d \times r$ matrix whose columns have norm at most one. Put $P = VV^* \succeq 0$ and $M = V^*V$, and let Q be Hermitian with $n_+(Q) \leq b$. Then*

$$\|P + Q\|_{\text{F}}^2 \geq 4 \text{tr}(P + Q) - 3r - 4b + \mathcal{D}(M). \quad (4)$$

Proof. Write $Q = Q_+ - Q_-$ with $Q_\pm \succeq 0$, $Q_+ Q_- = 0$, and $\text{rank } Q_+ \leq b$. Since $P, Q_+ \succeq 0$,

$$\|P + Q\|_{\text{F}}^2 \geq \|P - Q_-\|_{\text{F}}^2 + \|Q_+\|_{\text{F}}^2.$$

If $q_j > 0$ are the nonzero eigenvalues of Q_+ , then $q_j^2 \geq 4q_j - 4$, so

$$\|Q_+\|_{\text{F}}^2 \geq 4 \text{tr } Q_+ - 4b. \quad (5)$$

Let $p_1 \geq \dots \geq p_r \geq 0$ be the eigenvalues of M (the nonzero eigenvalues of P , padded by zeros), and let $n_1 \geq n_2 \geq \dots \geq 0$ be the eigenvalues of Q_- . Von Neumann's trace inequality gives $\text{tr}(PQ_-) \leq \sum_{i \leq r} p_i n_i$, hence

$$\|P - Q_-\|_{\text{F}}^2 + 4 \text{tr } Q_- \geq \sum_{i \leq r} ((p_i - n_i)^2 + 4n_i).$$

For $p \geq 0$,

$$\min_{n \geq 0} ((p - n)^2 + 4n) = \begin{cases} p^2, & 0 \leq p \leq 2, \\ 4p - 4, & p \geq 2, \end{cases} = 2p - 1 + \Psi(p).$$

Therefore

$$\|P - Q_-\|_F^2 \geq 2 \operatorname{tr} P - r + \mathcal{D}(M) - 4 \operatorname{tr} Q_-. \quad (6)$$

Combining (5) and (6) gives

$$\|P + Q\|_F^2 \geq 4 \operatorname{tr}(P + Q) - 2 \operatorname{tr} P - r - 4b + \mathcal{D}(M).$$

Since $\operatorname{tr} P$ is the sum of the squared column norms of V , $\operatorname{tr} P \leq r$. $\square$

Corollary 2.2. *With $M := V^\top V$,*

$$s_1 \geq 4 \operatorname{tr} \hat{A} - \|\hat{A}\|_F^2 - 2N(I') + \mathcal{D}(M). \quad (7)$$

After removing the tails as in Proposition 4.2 of [1],

$$N_0^s(T, 2T) \geq H_{\text{MT}} N(T, 2T) + \mathcal{D}(M^\circ) - o(N(T, 2T)), \quad (8)$$

where M° is the Gram matrix of the retained central simple zeros.

Proof. Apply Lemma 2.1 to $P_1 + Q' = \hat{A}$. Equations (1) and (2) give

$$\|\hat{A}\|_F^2 \geq 4 \operatorname{tr} \hat{A} - 3s_1 - 4s_2 - 4p + \mathcal{D}(M) \geq 4 \operatorname{tr} \hat{A} - s_1 - 2N(I') + \mathcal{D}(M),$$

which is (7).

Use the same comparison $\hat{A} = \hat{G} - \hat{E}$ and tail estimates as in Propositions 4.2 and 4.4 of [1]. Pinching the full Gram matrix into the retained central block and its complement gives $\mathcal{D}(M) \geq \mathcal{D}(M^\circ)$ because $X \mapsto \operatorname{tr} \Psi(X)$ is convex and unitarily invariant. Moreover $N(I') = N(T, 2T) + o(N(T, 2T))$, and Theorem D gives

$$\operatorname{tr} \hat{G} = N(T, 2T)(1 + o(1)), \quad \|\hat{G}\|_F^2 = \left(\frac{1}{c_1^*} + o(1) \right) N(T, 2T),$$

where $2 - 1/c_1^* = H_{\text{MT}}$. Substitution in (7) yields (8). $\square$

3 The Montgomery–Taylor overlap kernel

Put $h := 2\pi/L$ and define the normalised ordinate

$$x_\rho := \frac{\gamma_\rho - T}{h} = \frac{L(\gamma_\rho - T)}{2\pi}.$$

Lemma 3.1. *Fix $R_0 > 0$. Delete the simple zeros in strips of normalised width L^2 at the two ends of the grid. The number deleted is $o(N(T, 2T))$, and uniformly for retained simple zeros with $|x_\rho - x_{\rho'}| \leq R_0$,*

$$\langle v_\rho, v_{\rho'} \rangle = k(x_\rho - x_{\rho'}) + o(1), \quad (9)$$

where

$$k(x) := \frac{K(x)}{K(0)}, \quad (10)$$

$$\begin{aligned} K(x) &:= \int_{-1/2}^{1/2} \cos(\sqrt{2}t) \cos(2\pi xt) dt \\ &= \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}}, \end{aligned} \quad (11)$$

and $K(0) = \sqrt{2} \sin(1/\sqrt{2})$. The singularities in (11) are removable.

Proof. For the full Gabor grid, Lemma 2.2 of [1] gives

$$\frac{1}{aL^2} \sum_{j \in \mathbb{Z}} \widehat{\phi}(\gamma - \tau_j) \widehat{\phi}(\gamma' - \tau_j) = \frac{\Phi(\gamma - \gamma')}{aL}, \quad \Phi = \widehat{\phi^2}.$$

For points at normalised distance at least L^2 from the grid ends, the terms with $j \notin [0, d]$ contribute $O(L^{-2})$, uniformly when $|x_\rho - x_{\rho'}|$ is bounded, by the r^{-2} decay used in [1, §5.3].

For fixed x , substitution $u = Lt$ gives the normalised full-grid overlap

$$\frac{\Phi(hx)}{aL} = \frac{\int_{-1/2}^{1/2} \phi(Lt)^2 e^{-2\pi i x t} dt}{\int_{-1/2}^{1/2} \phi(Lt)^2 dt} \rightarrow \frac{K(x)}{K(0)}. \quad (12)$$

Indeed, $\phi(Lt)^2 \rightarrow \cos(\sqrt{2}t) \mathbf{1}_{[-1/2, 1/2]}(t)$ in L^1 ; the fixed-width taper has rescaled length $O(L^{-1})$. The convergence in (12) is uniform for x in compact sets. Together with the tail estimate, this proves (9). Finally, the deleted strips have ordinate length $O(L)$ and contain $O(L^2) = o(N(T, 2T))$ zeros, since $N(t+1) - N(t) \ll \log(t+3)$. $\square$

Set

$$w(x) := k(x)^2.$$

4 A seven-point local inequality

For nonnegative gaps $g_1, \dots, g_6$, define

$$\mathcal{F}_6(g_1, \dots, g_6) := \frac{1}{3000} \sum_{i=1}^6 g_i + \sum_{r=1}^6 \frac{2}{7-r} \sum_{i=1}^{7-r} w(g_i + \dots + g_{i+r-1}). \quad (13)$$

The second term contains all 21 pairwise separations among seven ordered points.

Proposition 4.1 (Certified local inequality). *For every $g_1, \dots, g_6 \geq 0$,*

$$\mathcal{F}_6(g_1, \dots, g_6) \geq \beta_0 := \frac{3,826,217}{10^9}. \quad (14)$$

Verification. The verifier at

<https://github.com/npip99/zeta-zeros>

uses Arb interval arithmetic through `python-flint`. It evaluates the entire-sinc expression

$$K(x) = \frac{1}{2} \left(\frac{\sin(\pi x - 1/\sqrt{2})}{\pi x - 1/\sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{\pi x + 1/\sqrt{2}} \right)$$

at 128-bit precision and constructs rigorous lower bounds for w on a grid of mesh $1/8000$. If $\sum g_i \geq 3000\beta_0 = 11.478651$, the linear term proves the claim. On the remaining compact region, the one-variable term

$$U(g) := \frac{g}{3000} + \frac{1}{3}w(g)$$

reduces each coordinate to three certified unions of grid cells. The resulting 3^6 boxes are proved by interval lower bounds, a certified convex tangent bound when applicable, or bisection of the widest coordinate. The program fails if a terminal box remains unresolved.

The committed certificate records `verified=true`, grid 8000, precision 128 bits, 963,245 visited nodes, and maximum depth 69. The adjacent target $3,826,218/10^9$ fails at a terminal cell, so the certified target is maximal on the 10^{-9} lattice at this grid. It also records the hashes of the interval tables used in the search. $\square$

Lemma 4.2. *Let $y_1 < \dots < y_m$ with $m \geq 7$, and put*

$$E_m := 2 \sum_{1 \leq i < j \leq m} w(y_j - y_i).$$

Then

$$E_m + \frac{1}{500}(y_m - y_1) \geq \beta_0(m - 6). \quad (15)$$

Proof. Write $g_i = y_{i+1} - y_i$ and sum (14) over the $m - 6$ consecutive seven-point windows. A pair spanning r gaps occurs at most $7 - r$ times and has coefficient $2/(7 - r)$ each time, so its total contribution is at most $2w(y_j - y_i)$. Each single gap occurs at most six times, giving at most

$$\frac{6}{3000} \sum_{i=1}^{m-1} g_i = \frac{1}{500}(y_m - y_1).$$

$\square$

Lemma 4.3. *If $G \succeq 0$ is Hermitian, then*

$$\mathrm{tr} \Psi(G) \geq \min \left\{ 1, 2 \sum_{i < j} |G_{ij}|^2 \right\}. \quad (16)$$

Proof. If every eigenvalue of G is at most 2, then $\Psi(G) = (G - I)^2$, and

$$\mathrm{tr} \Psi(G) = \mathrm{tr}(G - I)^2 \geq 2 \sum_{i < j} |G_{ij}|^2.$$

If some eigenvalue $\lambda > 2$, then $\Psi(\lambda) = 2\lambda - 3 > 1$. $\square$

Take

$$m := 267, \quad A_0 := \beta_0(m - 6) = \frac{998,642,637}{10^9} < 1. \quad (17)$$

Let B be a block of 267 consecutive retained simple zeros. Write G_B for its Gram matrix. If $\mathrm{span}(B)/500 \geq A_0$, then (18) below is immediate. Otherwise $\mathrm{span}(B) < 500$, so Lemma 3.1 applies uniformly to every pair in B . Since m is fixed,

$$2 \sum_{i < j} |(G_B)_{ij}|^2 = E_m + o(1).$$

Lemmas 4.2 and 4.3 give, uniformly in B ,

$$\mathcal{D}(G_B) + \frac{1}{500} \mathrm{span}(B) \geq A_0 - o(1). \quad (18)$$

5 Shifted block pinching

Let

$$x_1 < \cdots < x_{S^\circ}$$

be the normalised ordinates of the retained central simple zeros. Then

$$S^\circ = N_0^s(T, 2T) - o(N(T, 2T)). \quad (19)$$

For each of the $m = 267$ offsets, partition the ordered points into full consecutive blocks of size m , leaving at most $2(m - 1)$ endpoint points. Pinching M° to the full principal blocks and the remaining block gives

$$\mathcal{D}(M^\circ) \geq \sum_B \mathcal{D}(G_B). \quad (20)$$

This follows because pinching is an average of unitary conjugations and $X \mapsto \text{tr } \Psi(X)$ is convex and unitarily invariant.

For each offset, sum (18) over the full blocks and use (20); then average over the m offsets. The average number of full blocks is $S^\circ/m + O(1)$. Each gap $x_{j+1} - x_j$ belongs to a block span for at most $m - 1$ offsets, and

$$x_{S^\circ} - x_1 \leq \frac{T}{h} = \frac{LT}{2\pi} = d + O(1) = N(T, 2T) + o(N(T, 2T))$$

by the Riemann–von Mangoldt formula. The blockwise $o(1)$ term in (18) is uniform, so its total is $o(N(T, 2T))$. Therefore

$$\mathcal{D}(M^\circ) \geq \frac{A_0}{m} N_0^s(T, 2T) - \frac{m-1}{500m} N(T, 2T) - o(N(T, 2T)). \quad (21)$$

With $m = 267$ and $A_0 = 998,642,637/10^9$,

$$\mathcal{D}(M^\circ) \geq \frac{332,880,879}{89 \cdot 10^9} N_0^s(T, 2T) - \frac{133}{66,750} N(T, 2T) - o(N(T, 2T)). \quad (22)$$

Substitute (22) into (8). This gives

$$\begin{aligned} N_0^s(T, 2T) &\geq H_{\text{MT}} N(T, 2T) + \frac{332,880,879}{89 \cdot 10^9} N_0^s(T, 2T) \\ &\quad - \frac{133}{66,750} N(T, 2T) - o(N(T, 2T)). \end{aligned}$$

Hence

$$\left(1 - \frac{332,880,879}{89 \cdot 10^9}\right) N_0^s(T, 2T) \geq \left(H_{\text{MT}} - \frac{133}{66,750}\right) N(T, 2T) - o(N(T, 2T)).$$

Divide by $N(T, 2T)$ and let $T \rightarrow \infty$ to obtain

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{267 \cdot 10^9 H_{\text{MT}} - 5.32 \cdot 10^8}{266,001,357,363} = 0.6730254674531403414 \dots$$

This proves Theorem 1.1.

Reproducibility

Proposition 4.1 is the only computer-assisted input. Its source, tests, trust-base description, and certificate are available at

<https://github.com/npip99/zeta-zeros>.

The verifier reconstructs every transcendental interval enclosure from the stated formulas.

References

- [1] Claude, *More than two thirds of the zeros of the Riemann zeta function lie on the critical line*, preprint, August 10, 2026, available online.
- [2] ainta, *zeta-simple-zeros: a 67.30085% lower bound for simple zeros of the Riemann zeta function*, research artifact, August 10, 2026, available online. The argument and verifier of the present note originate there; only the certified target in Proposition 4.1 and the propagated constants differ.
- [3] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), no. 8, 1281–1292.
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